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B.TECH. (CSE)
V SEMESTER
UE24CS341A – SOFTWARE ENGINEERING

PRODUCT BACKLOG

ON

**UpSkill-AI: An AI-Powered Upskilling Recommendation
and Automation System for Students**

Agile Project Management Deliverable (Jira)

TEAM NUMBER: 18 PROJECT ID: 22

SUBMITTED BY

Name	SRN	Module Owned
VANSH SHARMA	PES2UG24AM179	M1 – Profile & Skill Extraction
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Revision History

Name	Date	Reason For Changes	Version
VANSH SHARMA, YAJAT SONI, HEMANTH KUMAR R, SHUBHIKA PRADEEP	30-09-2026	Initial version	1.0

1. Introduction

1.1 Purpose

This document is the product backlog for UpSkill-AI. It records the epics and user stories that make up the project, their acceptance criteria, effort estimates, priorities, sprint allocation and ownership, and it traces every story back to the requirements it satisfies.

The backlog is maintained in Jira. This document is the exported and reviewed snapshot submitted as the Phase-2 agile project management deliverable; Jira remains the working source of truth and will continue to change as the project progresses.

1.2 How the Board is Organised

Field	Convention used
Project key	UPSK
Issue types	Epic, Story, Task, Bug
Board type	Scrum, with a sprint backlog drawn from this product backlog
Story key prefixes	UA (foundation), UP (profile), RM (roadmap), PR (progress), CA (catalogue and access)
Estimation	Story points on a modified Fibonacci scale: 1, 2, 3, 5, 8, 13
Sprint length	Two weeks, five sprints (S1 to S5)
Labels	module:M1 M2 M3 M4, plus type labels such as security, external-api, seed-data
Linking	Every story carries the requirement IDs it satisfies, which gives two-way traceability with the SRS and the test plan

1.3 Workflow States

State	Meaning	Exit condition
Backlog	Accepted into the product backlog but not scheduled	Assigned to a sprint
Selected for Development	In the current sprint, not yet started	A developer picks it up
In Progress	Being implemented on a feature branch	Implementation complete and unit tests written
In Review	Pull request open	Approved by at least one other member and CI green
In Test	Merged; test cases being executed	All linked test cases pass
Done	Meets the Definition of Done	—

1.4 Definition of Done

A story is Done only when all of the following hold. This is deliberately strict because unit test coverage and static analysis are separately assessed.

- Acceptance criteria met and demonstrated to at least one other team member
- Code merged to main through a reviewed and approved pull request
- Unit tests written for the new logic and passing in CI
- Linked test cases from the Project Test Plan executed and passing
- SonarQube quality gate passed with no new blocker or critical issue
- No cross-module internal imports introduced (architecture rules R1 and R2)
- Any change to a shared contract agreed by all four owners (rule R3)
- Requirement IDs on the ticket verified as genuinely satisfied, not merely touched

1.5 Estimation Basis

Story points express relative effort and uncertainty, not hours. The reference points the team calibrated against are:

Points	Meaning	Reference story
1–2	Trivial or purely mechanical; no unknowns	UA-6 freeze shared contracts
3	Well understood; a few hours of focused work	UP-2 create and view profile
5	Standard feature with some integration work	UP-11 compute and expose the skill vector
8	Significant logic or a genuine unknown	RM-5 sequence by prerequisite
13	Too large; must be split before it enters a sprint	None currently in the backlog

2. Epics

Epic key	Epic	Goal	Owner	Items	Points
UPSK-E0	Foundation and Delivery Pipeline	Repository, containerisation, CI/CD, migrations, shared contracts and seed data — everything the other four epics depend on.	All members	8	36
UPSK-E1	Student Profile and Skill Extraction	Capture what a student already knows and express it as a skill vector.	Vansh Sharma (M1)	14	64
UPSK-E2	Skill Gap Analysis and Roadmap	Turn a skill vector and a target role into an ordered, explainable roadmap.	Yajat Soni (M2)	9	48
UPSK-E3	Progress Tracking and Adaptive Adjustment	Track what the student actually completes and adapt recommendations accordingly.	Hemanth Kumar R (M3)	9	45
UPSK-E4	Resource Catalogue, Authentication and Access	Provide the curated resource catalogue and enforce who may do what.	Shubhika Pradeep (M4)	11	50
	Total			51	243

Total committed scope: 243 story points across 5 epics and 51 backlog items.

Epic E0 is deliberately owned by the whole team rather than one member. It contains the work that every other epic depends on — particularly taxonomy and catalogue seeding, which has no natural home in the module split and would otherwise be nobody's job until it blocked everyone.

3. Backlog — Epic E0: Foundation and Delivery Pipeline

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
UA-1	Story	Repository and branch setup	As a developer, I want a GitHub repository with a documented branching convention so that four people can work in parallel without conflicts.	Repo created; main protected; feature/<module>-<ticket> convention documented in README; PR review required before merge.	3	Highest	S1	All	Constraint C2
UA-2	Story	Docker Compose stack	As a developer, I want the full stack to start with one command so that every environment behaves identically.	docker-compose.yml defines upskill-web, upskill-api, upskill-db and the pgdata volume; stack starts clean on a fresh machine.	5	Highest	S1	M4	Goal G5, C2
UA-3	Story	Jenkins pipeline skeleton	As a developer, I want CI to run on every pull request so that regressions are caught before merge.	Pipeline runs checkout, install, lint, pytest with coverage, and publishes the coverage report; failing tests block the merge.	5	Highest	S1	M3	Constraint C2
UA-4	Story	SonarQube quality gate	As a team, we want static analysis wired into CI so that code quality is measured continuously.	SonarQube scan runs in the pipeline; a failing quality gate blocks promotion of the build.	3	High	S1	M3	Constraint C2
UA-5	Story	Database schema and migrations	As a developer, I want versioned migrations so that four people can change the schema in parallel safely.	Alembic configured; initial migration creates all tables from the ER model; migration runs automatically on container start.	5	Highest	S1	M4	SRS §4.6
UA-6	Task	Freeze shared contracts	As a team, we want the skill vector format and taxonomy schema frozen before implementation so that module interfaces do not shift mid-sprint.	Skill vector format and taxonomy schema documented in Appendix B and signed off by all four owners; change requires unanimous agreement.	2	Highest	S1	All	Arch rule R3
UA-7	Task	Seed skill taxonomy and role requirements	As a team, we want seeded reference data so that gap computation returns meaningful output during development and demonstration.	At least 40 canonical skills with levels, aliases and prerequisite edges; at least 2 target roles with required skills and levels.	8	Highest	S1	All	Risk A1
UA-8	Task	Seed the resource catalogue	As a team, we want a seeded catalogue so that roadmaps contain real resources rather than empty results.	At least 25 approved resources across the seeded skills, each tagged with skill, level and duration.	5	High	S3	All	Risk A1, ADR-03

4. Backlog — Epic E1: Student Profile and Skill Extraction

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
UP-1	Story	Student registration and login	As a student, I want to register and log in so that my profile is private to me.	Registration validates email format and password strength; login issues a JWT; invalid credentials return a generic error without indicating which field failed.	5	Highest	S1	M1/M4	REQ-F4-01, F4-02
UP-2	Story	Create and view profile	As a student, I want to create a profile so that the system knows what I have already studied.	Exactly one profile per account; a second create attempt is rejected; student can view only their own profile.	3	Highest	S1	M1	REQ-F1-01, F1-39
UP-3	Story	Manage course entries	As a student, I want to add my completed courses so that the skills they imply are captured.	Add, edit and delete courses; mandatory fields validated; grade outside the permitted set is rejected with the field identified.	5	High	S2	M1	REQ-F1-02, F1-35, F1-36
UP-4	Story	Manage project entries	As a student, I want to record my projects so that practical skills are reflected in my profile.	Add, edit and delete projects; description capped at 2000 characters with a live character count; optional repository URL validated for format.	3	High	S2	M1	REQ-F1-03, F1-37
UP-5	Story	Manually add a skill	As a student, I want to declare a skill directly so that I can correct anything the system missed.	Skill must be selected from the taxonomy; an unknown skill is rejected with the closest canonical matches shown as suggestions.	3	High	S2	M1	REQ-F1-04, F1-05
UP-6	Story	Upload a resume	As a student, I want to upload my resume so that I do not have to enter every skill by hand.	PDF and DOCX only, rejected otherwise with the exact message; files over 5 MB rejected; processing indicator shown; file deleted 30 days after processing.	8	High	S2	M1	REQ-F1-08 to F1-11, F1-18
UP-7	Story	Extract skills from a resume	As a student, I want skills detected from my resume so that profile setup is fast.	Candidate skills identified with a confidence measure and the source sentence; an unreadable resume offers manual entry instead of an error.	8	High	S2	M1	REQ-F1-12 to F1-14
UP-8	Story	Confirm extracted skills	As a student, I want to review detected skills before they are saved so that wrong extractions never enter my profile.	No extracted skill is stored without explicit confirmation; each can be confirmed, level-edited or rejected individually; bulk confirm available above the high-confidence threshold.	5	Highest	S2	M1	REQ-F1-15 to F1-17
UP-9	Story	Normalise skill variants	As a student, I want different spellings of the same skill treated as one so that my profile is not cluttered with duplicates.	Known variants resolve to one canonical skill; matching ignores case and whitespace; no duplicate entries created.	5	Highest	S2	M1	REQ-F1-19 to F1-21, F1-24
UP-10	Story	Handle unresolved skills	As an administrator, I want unrecognised skill strings surfaced so that the taxonomy can be extended over time.	Unresolvable strings are recorded, never silently dropped; the student is asked to map them; administrators can review the full list.	3	Medium	S3	M1	REQ-F1-22, F1-23
UP-11	Story	Compute and expose the skill vector	As the roadmap module, I need the student's skill vector so that I can compute their gap.	Vector recomputed on every profile change; highest level wins when sources conflict; exposed in the frozen Appendix B format.	5	Highest	S3	M1	REQ-F1-25 to F1-28
UP-12	Story	Accept skill vector updates	As the progress module, I need to raise a skill level after a completion so that the profile stays current.	Level raised only if the incoming level is higher; a lower or equal level is a no-op; change written to history with its source.	5	Highest	S4	M1	REQ-F1-29, ADR-04
UP-13	Story	Select a target role	As a student, I want to choose a career goal so that recommendations are aimed at something specific.	One active role at a time; changing it marks existing recommendations stale; a summary shows required, held and gap counts.	3	Highest	S3	M1	REQ-F1-31 to F1-34
UP-14	Story	Skill change history	As a student, I want to see how my recorded skills changed so that the	Every change records skill, previous level, new level, source and timestamp.	3	Medium	S5	M1	REQ-F1-06, F1-07

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
			profile is explainable.						

5. Backlog — Epic E2: Skill Gap Analysis and Roadmap

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
RM-1	Story	Compute the skill gap	As a student, I want to see exactly which skills I am missing so that I know what to work on.	Gap includes skills absent from the vector and skills held below the required level; computed as a pure function with no I/O.	8	Highest	S3	M2	REQ-F2-01, F2-02
RM-2	Story	Rank gap by relevance	As a student, I want the most important gaps first so that I do not waste time on peripheral skills.	Gap entries ordered by relevance and importance to the target role; order is deterministic for identical input.	5	High	S3	M2	REQ-F2-03
RM-3	Story	Display the prioritised gap	As a student, I want the gap shown clearly so that I can understand it at a glance.	Gap rendered as a prioritised list showing current level against required level for each skill.	3	High	S3	M2	REQ-F2-04
RM-4	Story	Recommend resources per gap skill	As a student, I want specific resources for each missing skill so that I can act on the analysis.	Each gap skill returns ranked catalogue resources covering that skill at the appropriate level.	8	Highest	S4	M2	REQ-F2-05
RM-5	Story	Sequence by prerequisite	As a student, I want the roadmap ordered so that I am never recommended an advanced item before its prerequisite.	A resource covering a prerequisite skill always precedes a resource that depends on it; a cyclic definition is detected and reported rather than looping.	8	Highest	S4	M2	REQ-F2-06
RM-6	Story	Explain each recommendation	As a student, I want to know why something was recommended so that I trust the system.	Every roadmap item displays the specific skill gap it closes.	3	Highest	S4	M2	REQ-F2-07, Goal G3
RM-7	Story	Recompute on change	As a student, I want my roadmap to reflect my current profile so that it never contradicts work I have just done.	Gap and roadmap recomputed whenever the skill vector or target role changes; last recalculation timestamp recorded.	5	Highest	S4	M2	REQ-F2-08, F2-11, ADR-05
RM-8	Story	External fallback for uncovered skills	As a student, I want a suggestion even when the catalogue has nothing for a skill so that no gap is left blank.	When the catalogue has no resource, the external API or a cached fallback list is used; the student sees no error and the rest of the roadmap is unaffected.	5	High	S5	M2/M4	REQ-F2-09, ADR-03
RM-9	Story	Cache external lookups	As a team, we want external results cached per skill so that the free-tier quota is not exhausted.	Results cached per canonical skill with a configurable TTL; a repeat request within the TTL makes no external call.	3	High	S4	M4	REQ-F2-10, Constraint C4

6. Backlog — Epic E3: Progress Tracking and Adaptive Adjustment

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
PR-1	Story	Enrol in a resource	As a student, I want to enrol in a recommended resource so that my progress can be tracked.	Progress record created with initial state; enrolment against an invalid resource is rejected.	3	High	S3	M3	REQ-F3-01, F3-16
PR-2	Story	Update progress	As a student, I want to record how far through a resource I am so that my dashboard is accurate.	Completion percentage accepted in the 0 to 100 range; out-of-range values rejected with a validation message.	3	High	S3	M3	REQ-F3-02, F3-03
PR-3	Story	Mark a resource complete	As a student, I want to mark something finished so that my skill profile updates.	Completion recorded; evidence requested where the resource requires it; completion against a non-enrolled resource is blocked.	5	Highest	S4	M3	REQ-F3-04, F3-16
PR-4	Story	Submit completion evidence	As a student, I want to upload a certificate or quiz score so that my claimed completion is credible.	Certificate metadata or quiz score stored and linked to the progress record; invalid or missing evidence blocks evidence-based completion; scores validated for range.	8	Highest	S4	M3	REQ-F3-05 to F3-07
PR-5	Story	Update the skill vector on valid evidence	As a student, I want completing a course to raise my recorded skill so that the system adapts.	On validated evidence, M3 calls M1 to raise the affected skill levels; M3 never writes the vector directly.	5	Highest	S5	M3	REQ-F3-08, ADR-04
PR-6	Story	Trigger adaptive re-ranking	As a student, I want my roadmap refreshed as soon as I complete something so that it is never out of date.	A validated vector change marks recommendations stale; the refreshed roadmap is returned on the next read without requiring re-login.	5	Highest	S5	M3	REQ-F3-09, F3-17, ADR-05
PR-7	Story	Progress dashboard	As a student, I want to see my progress over time so that I stay motivated.	Dashboard shows progress per enrolled resource, proportion of gaps closed, streaks and milestones.	8	High	S5	M3	REQ-F3-10 to F3-12
PR-8	Story	Detect stalled learning paths	As a student, I want a nudge when I stop making progress so that I do not silently drop off.	Paths with no qualifying progress within the configured interval are flagged; a reminder is generated per the notification rules.	5	Medium	S5	M3	REQ-F3-13, F3-14
PR-9	Story	Preserve progress history	As a student, I want my completed work to survive a roadmap change so that nothing I did is lost.	Progress records remain intact and correctly valued after recommendations are re-ranked or replaced.	3	High	S5	M3	REQ-F3-15

7. Backlog — Epic E4: Resource Catalogue, Authentication and Access

Key	Type	Summary	User story	Acceptance criteria	SP	Pri	Sprint	Owner	Requirement IDs
CA-1	Story	Role-based access control	As a team, we want permissions enforced centrally so that no endpoint is accidentally left unguarded.	A single authorize() check used by every module; students blocked from faculty and admin endpoints; faculty blocked from admin catalogue operations; enforcement is server-side.	8	Highest	S2	M4	REQ-F4-02 to F4-04, F4-18, NFR-SEC-02
CA-2	Story	Secure password storage	As a student, I want my password stored safely so that a database leak does not expose it.	Passwords hashed with Argon2id; plaintext never stored or written to logs.	3	Highest	S1	M4	REQ-F4-01, NFR-SEC-03
CA-3	Story	Create catalogue resources	As an administrator, I want to add learning resources so that students receive curated recommendations.	Resource stores title, description, type, skill tags, level, duration and source URL; required fields validated; skill tags must be canonical skills.	5	High	S2	M4	REQ-F4-05 to F4-08
CA-4	Story	Search and filter the catalogue	As a student, I want to browse resources so that I can look beyond my roadmap.	Search matches title, description and skill tags; filters for skill, level, type and duration; results paginated.	5	Medium	S3	M4	REQ-F4-09, F4-10, NFR-PERF-04
CA-5	Story	Moderate catalogue resources	As an administrator, I want to approve resources before students see them so that the catalogue stays trustworthy.	Resources carry a moderation state; only approved resources are returned to students; inactive resources are excluded from search and new recommendations.	5	High	S3	M4	REQ-F4-11, F4-12, BR-08
CA-6	Story	Faculty mentee view	As a faculty mentor, I want to see my mentees' progress so that I can intervene early.	Faculty can view assigned mentees' progress, gap and recommendation status; access is scoped to assigned mentees only; profiles are read-only to faculty.	5	High	S4	M4	REQ-F4-13, F1-40, NFR-SEC-08
CA-7	Story	Endorse a recommendation	As a faculty mentor, I want to endorse a recommendation so that my mentee knows it is worth prioritising.	Endorsement recorded against the recommendation, attributed to the faculty user, and visible in the student's roadmap view.	3	Medium	S5	M4	REQ-F4-14
CA-8	Story	Override a recommendation	As a faculty mentor, I want to override a recommendation when I know better so that guidance reflects context the system lacks.	Override requires a stated reason; the original algorithmic recommendation is retained and remains auditable, never silently deleted.	5	Medium	S5	M4	REQ-F4-15, BR-07
CA-9	Story	Audit administrative and faculty actions	As an administrator, I want a record of who changed what so that decisions are accountable.	Audit record written for catalogue changes, endorsements and overrides, capturing actor, action, target and reason.	3	Medium	S5	M4	REQ-F4-16
CA-10	Story	Role-appropriate analytics	As a faculty mentor, I want group reports so that I can spot students who have stalled.	Analytics limited to data the caller's role may access; stalled mentees identifiable from the report.	5	Low	S5	M4	REQ-F4-17, BR-09
CA-11	Story	Sanitised error responses	As a team, we want errors that reveal nothing internal so that the system is not a source of information for an attacker.	Internal errors return a generic message with a correlation id; no stack trace, token or internal identifier reaches the client.	3	High	S3	M4	NFR-SEC-10

8. Sprint Plan

Five two-week sprints covering 243 story points. Team capacity is four members. The sprints below are balanced to between 42 and 51 points, against an assumed velocity of roughly 50 points per sprint. That assumption is unvalidated until Sprint 1 completes and must be corrected against measured velocity thereafter.

Sprint	Dates	Sprint goal	Points	Items	Backlog items
S1	Weeks 1–2	Working pipeline and seeded reference data. Nothing user-facing, but every later sprint depends on it.	42	10	UA-1, UA-2, UA-3, UA-4, UA-5, UA-6, UA-7, UP-1, UP-2, CA-2
S2	Weeks 3–4	A student can build a profile from courses, projects and a resume, with skills normalised to the taxonomy.	50	9	UP-3, UP-4, UP-5, UP-6, UP-7, UP-8, UP-9, CA-1, CA-3
S3	Weeks 5–6	The skill vector drives a computed, ranked gap and a first roadmap. First demonstrable end-to-end value.	51	12	UA-8, UP-10, UP-11, UP-13, RM-1, RM-2, RM-3, PR-1, PR-2, CA-4, CA-5, CA-11
S4	Weeks 7–8	The full roadmap: resource recommendation, prerequisite sequencing, explanation, and completion recording.	50	9	UP-12, RM-4, RM-5, RM-6, RM-7, RM-9, PR-3, PR-4, CA-6
S5	Weeks 9–10	Adaptive re-ranking, faculty oversight, notifications and hardening ahead of the final demonstration.	50	11	UP-14, RM-8, PR-5, PR-6, PR-7, PR-8, PR-9, CA-7, CA-8, CA-9, CA-10

8.1 Sprint Sequencing Rationale

- S1 delivers nothing a user can see, and that is intentional. Seeding the taxonomy (UA-7) is on the critical path: without it, gap computation returns empty results and nothing in any later sprint can be demonstrated.
- S2 front-loads M1 because M2, M3 and M4 all consume the skill vector. Any slip in M1 propagates to every other module, so it is scheduled early with slack behind it.
- S3 is the first sprint producing a demonstrable end-to-end journey: profile in, ranked gap out. If the project falls behind, this is the sprint to protect.
- S4 completes the roadmap — resource recommendation, prerequisite sequencing and the explanation of each item — which is the assessed differentiator of the product.
- S5 carries the adaptive re-ranking path (PR-5, PR-6) alongside faculty features and hardening. Placing the adaptive behaviour in the final sprint is the single largest schedule risk in this plan; it depends on PR-4, which cannot start earlier. If S4 slips, PR-5 and PR-6 must be protected ahead of the faculty stories CA-7 to CA-10.

8.2 Dependencies Between Stories

Blocked story	Blocked by	Nature of the dependency
RM-1 Compute the skill gap	UP-11 Compute and expose the skill vector; UA-7 Seed taxonomy	Cannot compute a gap without a vector to compare and role requirements to compare against
RM-4 Recommend resources per gap skill	CA-3 Create catalogue resources; UA-8 Seed catalogue	No resources to recommend until the catalogue exists and is populated
RM-5 Sequence by prerequisite	UA-7 Seed taxonomy (prerequisite edges)	Sequencing needs prerequisite relationships in the taxonomy
PR-5 Update vector on valid evidence	UP-12 Accept skill vector updates	M3 cannot write the vector; it depends on M1 exposing the update operation (ADR-04)
PR-6 Trigger adaptive re-ranking	RM-7 Recompute on change	Re-ranking requires the recompute path to exist
CA-6 Faculty mentee view	CA-1 Role-based access control	Mentee scoping is an authorisation rule and needs the central check in place
All UP, RM, PR, CA stories	UA-5 Database schema and migrations; UA-6 Freeze shared contracts	No feature work starts before the schema and shared contracts are settled

9. Backlog Summary

9.1 Distribution by Epic and Sprint

Epic	S1	S2	S3	S4	S5	Items	Points
E0 Foundation and Delivery Pipeline	31	—	5	—	—	8	36
E1 Student Profile and Skill Extraction	8	37	11	5	3	14	64
E2 Skill Gap Analysis and Roadmap	—	—	16	27	5	9	48
E3 Progress Tracking and Adaptation	—	—	6	13	26	9	45
E4 Catalogue, Authentication and Access	3	13	13	5	16	11	50
Total	42	50	51	50	50	51	243

Every figure in this table is derived from the backlog items in sections 3 to 7; the row and column totals reconcile exactly.

9.2 Distribution by Priority

Priority	Items	Points	Interpretation
Highest	24	125	Core path. The system does not function without these; none may be dropped.
High	19	86	Required for a complete, credible product; drop only under severe schedule pressure.
Medium	7	27	Valuable but deferrable. This is the first cut line if velocity falls below plan.
Low	1	5	Nice to have. Expected to be cut without material loss.
Total	51	243	

9.3 Scope Risk

Three observations that the team should act on rather than file away:

- 125 of 243 points — 51 percent of total scope — are marked Highest priority and cannot be dropped without the system ceasing to function. That is a large non-negotiable core for a five-sprint, four-member project and leaves little absorptive capacity.
- The deferrable tier is only 32 points (Medium and Low combined). If measured velocity after Sprint 1 falls materially below 50 points, cutting the entire deferrable tier recovers roughly 13 percent of scope and no more. Beyond that the team must cut into High-priority work, which means dropping a visible feature. That decision should be taken deliberately at the Sprint 1 retrospective, not discovered in Sprint 4.
- Sprint 1 carries 42 points that produce nothing demonstrable. This is the correct sequencing — the taxonomy must exist before anything else works — but it means the first visible progress arrives in Sprint 2 and the first end-to-end journey in Sprint 3. Any slip in Sprint 1 pushes both back directly and should be escalated immediately rather than absorbed.

9.4 Requirement Coverage

Every functional requirement group in the SRS maps to at least one backlog story, and every story carries its requirement IDs. Combined with the Test Case ID column in the Project Test Plan, this gives a complete chain from requirement to story to test case, which is what Appendix C of the SRS records.

Requirement range	Covered by
REQ-F1-01 to F1-40	UP-1 to UP-14
REQ-F2-01 to F2-11	RM-1 to RM-9
REQ-F3-01 to F3-17	PR-1 to PR-9
REQ-F4-01 to F4-18	CA-1 to CA-10, UP-1
NFR-SEC-02, SEC-03, SEC-08, SEC-10	CA-1, CA-2, CA-6, CA-11
NFR-PERF-04	CA-4

Requirement range	Covered by
Constraints C2, C4 and Goal G5	UA-1 to UA-5, RM-9